Birla Institute of Technology and Science, Pilani
EEE C424 / INSTR C313 Microelectronic Circuits
Compre Exam I Semester 2012-2013

Max. Marks=~~20~~25
Time:45 min. Part A (open book) Date: 12-12-2012

Note- Please note that there will be no partial Marking. Please write to the point answers.

Q1. What is input resistance Rin (looking into the gate) of a diode-connected MOS transistor. Now, what is output resistance Rout of this transistor with source degeneration resistor Rs?

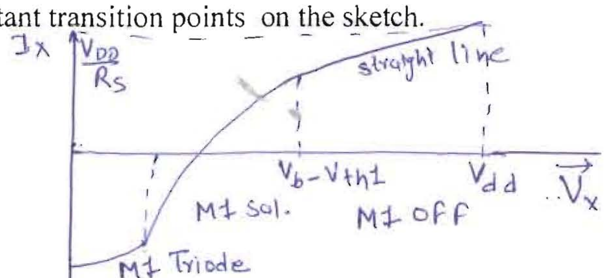
Ans. $R_{in} = 1/g_m$, $R_{out} = \frac{1}{g_m} + R_S$

Q2. When a common-drain stage is added between a common-source stage and a load resistor, what will be the effect on overall voltage gain and why?

Ans. Increase, CD stage isolates load register from CS stage and CD stage has low output resistance which increase drive capability and Gain increase.

Q3. Consider fig. Q_3, Sketch and label Ix versus Vx as Vx varies from 0 to Vdd. Identify important transition points on the sketch.

Ans.



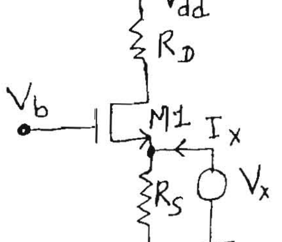


Fig. Q_3

Q4. Consider a bias voltage generator circuit shown in Fig. Q_4, Determine the value of V_2 and V_{dd} required for this circuit to operate. Given: $\lambda_n = \lambda_p = 0$, $V_{sb} = 0$, $I_{bias} = 200\mu A$, $(W/L)_1 = 5$, $(W/L)_2 = 3$, $(W/L)_3 = 11$, $\mu_n C_{ox} = 140 \mu A/V^2$, $V_{tn} = 0.7$ V, $\mu_p C_{ox} = 40 \mu A/V^2$, $V_{tp} = -0.8$ V

Ans.

$$V1 = 0.7 + \sqrt{\frac{2 \times 200}{140} \times \frac{1}{5}} = 1.46V$$

$$V2 = 1.46 + \sqrt{\frac{2 \times 200}{140} \times \frac{1}{3}} = 3.136$$

$$V_{dd} = 3.136 + \sqrt{\frac{2 \times 200}{40} \times \frac{1}{11}} = 4.89 V$$

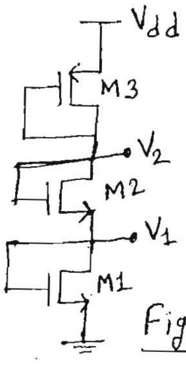


Fig. Q_4

Q5. Consider the circuit given in fig. Q_5, If the overall capacitance at node V_{out} is dominated by C_L=1pF and if r_o=200kΩ and g_m=600μA/V for all transistors, Determine the value of Rout & hence ω_{-3dB}?

Ans. $R_{out} = r_{o1} \| r_{o2} g_{m2} 1/g_{m3} \cong r_o/2 = 100k\Omega$

$$\omega_{-3dB} = \frac{2}{r_o C_L} \cong 10 \text{ Mrad/sec}$$

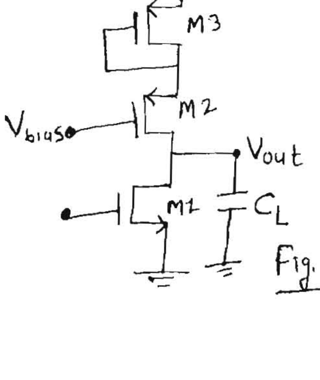


Fig. Q_5

Q6. Consider a circuit shown in Fig. Q_6, Determine the value of R_E and hence find out the maximum value of R_C for the BJT to remain in the active mode. Given: $\beta = 100$, $V_{BE} = 0.7V$, $I_C = 2mA$, $V_{CE(SAT)} = 0.2V$.

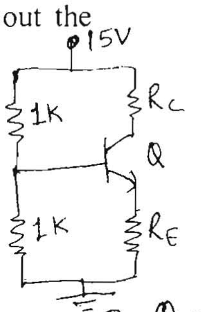


Fig Qc

Ans. $7.5 = 0.5k \times \frac{2m}{100} + 0.7 + (2m) R_E \Rightarrow \boxed{R_E = 3.4k}$

$15 - R_C(2m) - 2m(3.4k) < 0.2$

$\boxed{R_C \leq 4K}$

Q7.Identify type of feedback for circuit shown in Fig. Q_7.

Ans. Voltage sensing Current Mixing

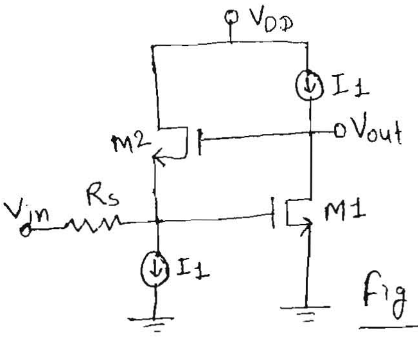


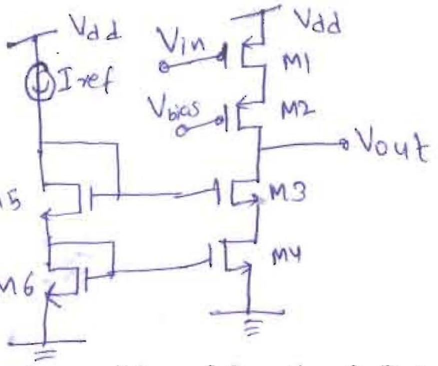
Fig Q_7

Q8. An amplifier having forward gain of 1000 and two poles at 1MHz and 10MHz. calculate the phase margin of a unity gain feedback amplifier.

GBW = 100 MHz

$P.M = 180^\circ - \left(\tan^{-1} \frac{100}{1} - \tan^{-1} \frac{10}{1} \right) = 6.28^\circ$

Q9. Sketch and Label a pMOS cascode amplifier with a complete nMOS cascode current mirror load.Assume I_{ref} of the current mirror is ideal.



Q10. For the cascade amplifier of Question 9, Determine the value of overall output resistance Rout. Given $r_o = 200k\Omega$ and $g_m = 600\mu A/V$ for all transistors.

Ans.

$R_{out} = r_o^2 g \,||\, r^2 g = \frac{1}{2} (200k)^2 (600u) = 12 M\Omega$

Solutions

Q1 → b) Poles → 4

$\omega_3(I/P) = 8.8\times10^7$ rad/sec

$\omega_2(O/P) = 4.5\times10^7$ rad/sec

ω_1 (Inter) $= 2.09\times10^6$ rad/sec

ω_4 (Atts) $= 6.25\times10^{10}$ rad/sec

(4M)

Zeros → 3

$\omega_{z_1} = g_m/C_{gd} = 50\times10^9$ rad/sec

$\omega_{z_3} =$ "

$\omega_{z_2} = -g_m/g_s = -2.5\times10^9$ rad/sec

(3M)

a) $A_{v_1} = -3.84$ V/V

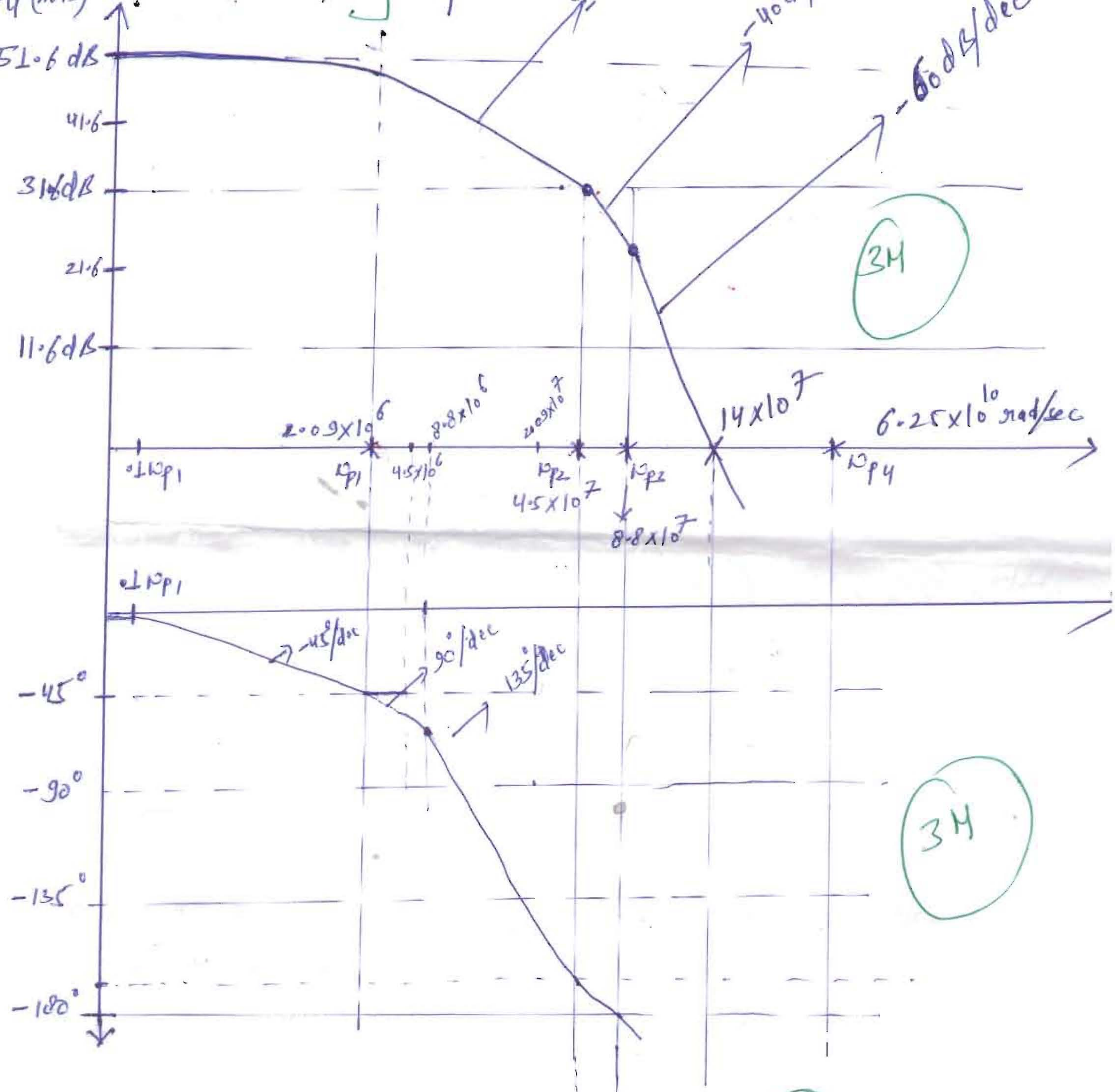
$A_{v_2} = -100$ V/V

$A_v = 384$

$A_v = 51.6$ d

(4M)

c)

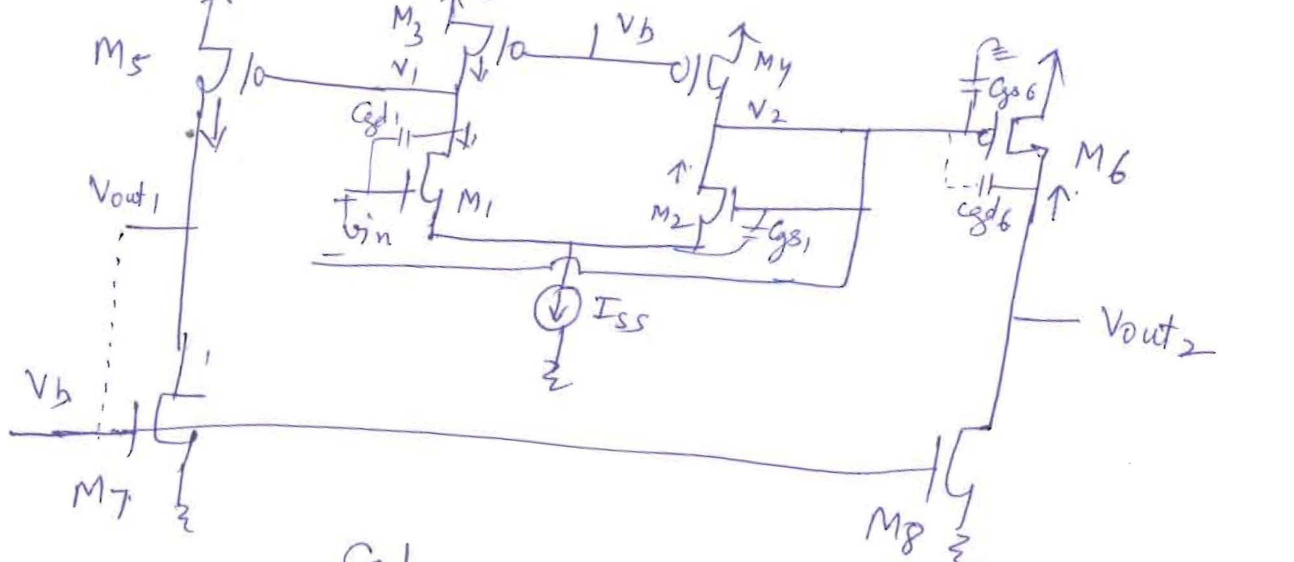


d) Phase Margin 0 — (2M)

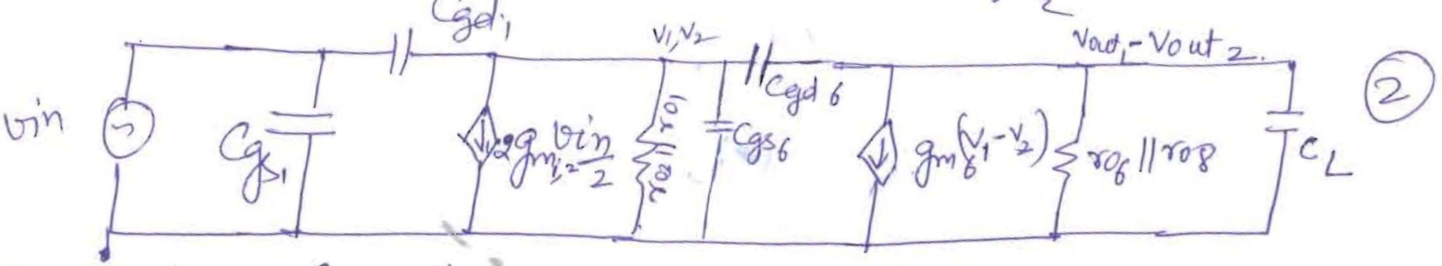
e) Plot will remain same. — (1M)

Q. 2

(a) High freq. Small Signal Model.



(a)



(2)

Without $V_{out_1}-M_7$ Connection $\rightarrow$

(b) Low freq. small signal model. Gain of $A_0 = v = +g_{m_1}(r_{o_1}\|r_{o_3})\,g_{m6}(r_{o_6}\|r_{o_8})$ (1)

(c) pole frequencies $\rightarrow$

ω_{p_1} = nodes $V_1 V_2 \longrightarrow \dfrac{-1}{(r_{o_1}\|r_{o_3})(C_{gs_5}+C_{gd_5}+\text{parasitic cap. } M_3, M_1, M_5)}$ (1)

ω_{p_2} = nodes $V_{out_1}, V_{out_2} \longrightarrow \dfrac{-1}{(r_{o_6}\|r_{o_8})(C_L + \text{para. } c \; M_5, M_7)}$ (1)

transmission Zero freq. -

ω_{z_1} = node $v_{in} \longrightarrow + g_{m_1}/C_{gd_1}$ (1/2)

ω_{z_2} node $V_{out_1}, V_{out_2} \longrightarrow + g_{m6}/C_{gd6}$. (1/2)

(d) Transfer function $\rightarrow$ $\dfrac{V_{out}(s)}{V_{in}(s)} = \dfrac{A_0\left(1-\frac{s}{\omega_{z_1}}\right)\left(1-\frac{s}{\omega_{z_2}}\right)}{\left(1+\frac{s}{|\omega_{p_1}|}\right)\left(1+\frac{s}{|\omega_{p_2}|}\right)}$ (1)

With $V_{out_1}-M_7$ Connection $\rightarrow$

(e) pole frequencies

node $V_1 V_2 = \omega_{p_1} = \dfrac{-1}{(r_{o_1}\|r_{o_3})(C_{gs_5} + \text{para. } c \; M_3, M_1, M_5)}$ (1)

pole frequency - ω_{p2} (node V_{out2}) $\simeq -\dfrac{1}{(r_{o6} \| r_{o8})(C_L + \text{para. of } M_6, M_8)}$ (1)

pole freq. ω_{p3} (node V_{out1}) $\simeq -\dfrac{1}{\left(\frac{1}{g_{m7}}\right)(C_{gs7} + C_{gs8} + C_{gd5} + C_{db5})}$ (1)

<u>Zero transmission frequency</u>

ω_{z1} (node v_{in}) $\simeq + g_{m1}/C_{gd1}$

ω_{z2} (node V_{out2}) $\simeq + g_{m6}/C_{gd6}$ (1)

ω_{z3} (node V_{out1}) $= -\dfrac{2 g_{m7}}{(C_L + C_{gs7} + C_{gs8} + C_{db5} + C_{gd5})}$ (1)

(f) Transfer function

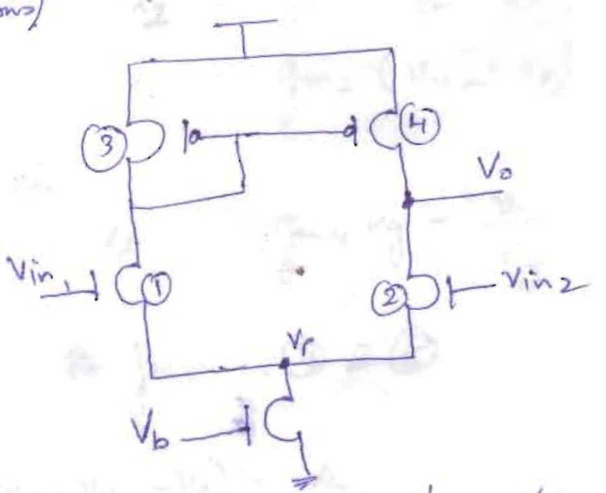
$$A(s) = \frac{V_{out2}(s)}{V_{in}(s)} \simeq \frac{A_0\left(1 - \frac{s}{\omega_{z1}}\right)\left(1 - \frac{s}{\omega_{z2}}\right)\left(1 + \frac{s}{2g_{m7}/(C_L + C_{gs8} + C_{gs7} + C_{gd5} + C_{db5})}\right)}{\left(1 + \frac{s}{|\omega_{p1}|}\right)\left(1 + \frac{s}{|\omega_{p2}|}\right)\left(1 + \frac{s}{|\omega_{p3}|}\right)}$$

(1) $\left(1 + \dfrac{s\,(C_L + C_{gs8} + C_{gs7} + \cdots)}{2 g_{m7}}\right)$

Q3

Solution Micro (Compre)

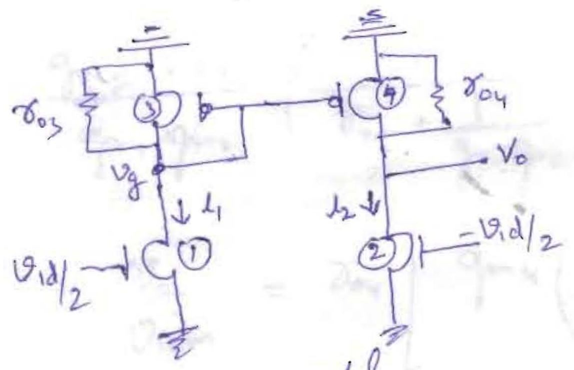
Ans)



Differential gain with $g_{m_1} \neq g_{m_2}$

$v_{in_1} - v_{in_2} = v_{id}$

$v_{in_1} = v_{id}/2$; $v_{in_2} = -v_{id}/2$



ac model

$$i_1 = g_{m_1} v_{id}/2 \quad ; \quad i_2 = g_{m_2}\left(\frac{-v_{id}}{2}\right) \quad -(1)$$

$$v_g = -i_1\left(\frac{1}{g_{m_3}} \parallel r_{o3}\right)$$

$$v_g = -g_{m_1}\left(\frac{v_{id}}{2}\right)\left(\frac{1}{g_{m_3}} \parallel r_{o3}\right) \quad -(2)$$

Also,

$$i_2 = -g_{m_4} v_g - \frac{v_o}{r_{o4}} \quad -(3)$$

Equation (1) & (3)

$$-g_{m_2}\frac{v_{id}}{2} = g_{m_4}\left(g_{m_1}\frac{v_{id}}{2}\right)\left(\frac{1}{g_{m_3}} \parallel r_{o3}\right) - \frac{V_o}{r_{o4}}$$

$$A_{dm} = \frac{v_o}{v_{id}}$$

$$\frac{v_o}{r_{o4}} = g_{m_4}\left(g_{m_1}\frac{v_{id}}{2}\right)\left(\frac{1}{g_{m_3}} \parallel r_{o3}\right) + g_{m_2}\frac{v_{id}}{2}$$

$$\frac{v_o}{v_{id}} = \frac{r_{o4}}{2}\left[g_{m_2} + g_{m_1} g_{m_4}\left(\frac{1}{g_{m_3}} \parallel r_{o3}\right)\right]$$

$$\frac{v_o}{v_{id}} = \frac{r_{o4}}{2}\left[g_{m_2} + g_{m_1}\left(\frac{r_{o3}}{\frac{1}{g_{m_3}} + r_{o3}}\right)\right]$$

$$\frac{v_o}{v_{id}} = \frac{r_{o4}}{2}\left[\frac{g_{m_2} r_{o3} + g_{m_1} r_{o3} + \frac{g_{m_2}}{g_{m_3}}}{\frac{1}{g_{m_3}} + r_{o3}}\right]$$

$$\frac{1}{g_{m_3}} \ll r_{o3}$$

$$\boxed{\frac{v_o}{v_{id}} = \frac{(g_{m_1} + g_{m_2})}{2} r_{o3} + \frac{g_{m_2}}{2 g_{m_3}}}$$

3

Putting values of $g_{m_1}, g_{m_2}, g_{m_3}$ & r_{o3}

$$A_{dm} = \left(\frac{1.001 + .999\,m}{2}\right) 100\,k\Omega + \frac{.999}{2 \times 1}$$

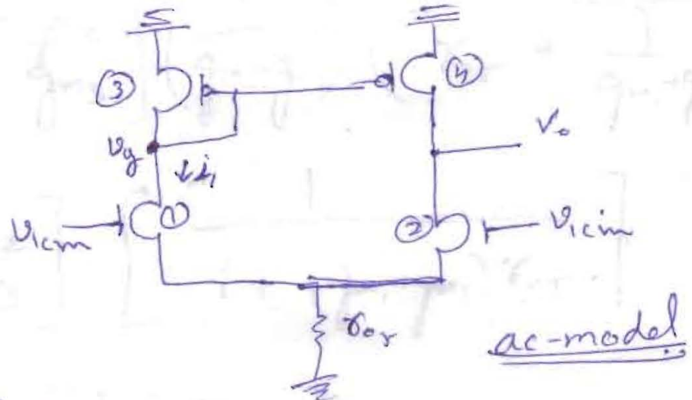
$$= 100.4995 \text{ V/V}$$

$$\approx 40 \text{ dB}$$

1

Common-mode gain

$v_{in_1} = v_{in_2} = v_{icm}$



ac-model

$$v_p = v_{icm}\frac{r_{o5}}{r_{o5} + \frac{1}{g_{m_1} + g_{m_2}}} \quad -(1)$$

$$v_g = -i_1\left(\frac{1}{g_{m_3}} \parallel r_{o3}\right) \quad (2)$$

Also,

$$i_2 = g_{m1}(v_{icm} - v_p) \quad -③$$

$$i_2 = g_{m2}(v_{icm} - v_p) \quad -④$$

Also,

$$i_2 = -g_{m4} v_g - \frac{v_o}{r_{o4}} \quad -⑤$$

from ④ & ⑤

$$g_{m2}(v_{icm} - v_p) = -g_{m4} v_g - \frac{v_o}{r_{o4}}$$

$$g_{m2}\left(v_{icm} - \frac{v_{icm} r_{o5}}{r_{o5} + \frac{1}{g_{m1}+g_{m2}}}\right) = g_{m4} g_{m1}\left(v_{icm} - \frac{v_{icm} r_{o5}}{r_{o5} + \frac{1}{g_{m1}+g_{m2}}}\right)\left(\frac{1}{g_{m3}} \parallel r_{o3}\right) - \frac{v_o}{r_{o4}}$$

$$\frac{g_{m2} v_{icm}}{g_{m1}+g_{m2}}\left(\frac{1}{r_{o5} + \frac{1}{g_{m1}+g_{m2}}}\right) = \frac{-v_o}{r_{o4}} + g_{m4}\left(\frac{1}{g_{m3}} \parallel r_{o3}\right)\left(\frac{g_{m1} v_{icm}}{(g_{m1}+g_{m2})}\right)\left(\frac{1}{r_{o5} + \frac{1}{g_{m1}+g_{m2}}}\right)$$

$$\frac{v_o}{v_{icm}} = r_{o4}\left[g_{m4}\left(\frac{1}{g_{m3}} \parallel r_{o3}\right)\left(\frac{g_{m1}}{g_{m1}+g_{m2}}\right)\left(\frac{1}{r_{o5} + \frac{1}{g_{m1}+g_{m2}}}\right) - \frac{g_{m2}}{g_{m1}+g_{m2}}\left(\frac{1}{r_{o5} + \frac{1}{g_{m1}+g_{m2}}}\right)\right]$$

$$\frac{v_o}{v_{icm}} = r_{o4}\left[\frac{r_{o3}}{\left(\frac{1}{g_{m3}} + r_{o3}\right)} g_{m1} - g_{m2}\right]\left(\frac{1}{g_{m1}+g_{m2}}\right)\left(\frac{1}{r_{o5} + \frac{1}{g_{m1}+g_{m2}}}\right)$$

$$= \frac{r_{o4}}{\left(\frac{1}{g_{m3}} + r_{o3}\right)}\left[(g_{m1} - g_{m2}) r_{o3} - \frac{g_{m2}}{g_{m3}}\right]\left(\frac{1}{g_{m1}+g_{m2}}\right)\left(\frac{1}{r_{o5} + \frac{1}{g_{m1}+g_{m2}}}\right)$$

$$= \left[(g_{m1} - g_{m2}) r_{o3} - \frac{g_{m2}}{g_{m3}}\right]\left[\frac{1}{1 + (g_{m1}+g_{m2}) r_{o5}}\right]$$

$$\boxed{A_{cm} = \frac{(g_{m1} - g_{m2}) r_{o3} - g_{m2}/g_{m3}}{1 + (g_{m1}+g_{m2}) r_{o5}}}$$

3

Putting values of g_m's & r_o's

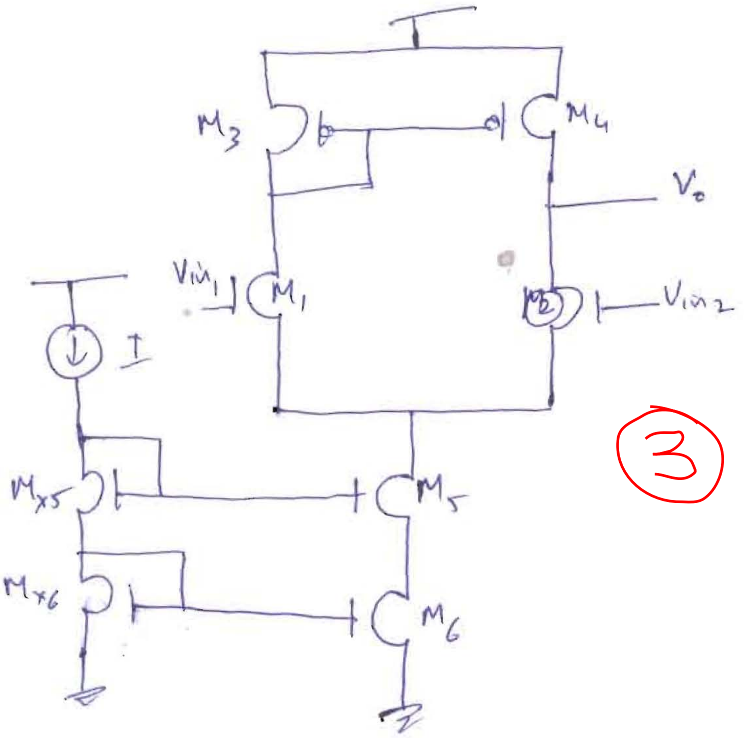
$$|A_{cm}| = 7.91 \times 10^{-3} \ V/V$$

$$= -42 \ dB \quad 1$$

$$CMRR = \frac{r_{o3}\left(\frac{g_{m1}+g_{m2}}{2}\right) + \frac{g_{m2}}{2g_{m3}}}{\frac{(g_{m1}-g_{m2})\,r_{o3} - g_{m2}/g_{m3}}{1+(g_{m1}+g_{m3})\,r_{o5}}} \quad 1$$

$$= 81.95 \ dB \quad 1$$

b) To increase CMRR, one has to decrease A_{cm}. to decrease A_{cm}, one has to increase the tail resistance source. 2



3

Q4

$I_D = \frac{1}{2}\mu_n C_{ox}(W/L)_n \times |V_{ov,n}|^2$

$200 = \frac{1}{2} \times 140 \times 71.5 \times |V_{ov,n}|^2$

$|V_{ov,n}| = 0.2\ V$ (1M)

similarly,

$I_D = \frac{1}{2}\mu_p C_{ox}(W/L)_p |V_{ov,p}|^2$

$200 = \frac{1}{2} \times 40 \times 250 \times |V_{ov,p}|^2$

$|V_{ov,p}| = 0.2\ V$ (1M)

(1M)

$g_{m1} = g_{m2} = g_{m3} = g_{m4} = \frac{2I_D}{V_{OV}} = 2\ mA/V$

$r_{o1} = r_{o2} = r_{o3} = r_{o4} = 25K$ (1M)

(i) $V_X = V_{in} - V_{GS4}$

For M_1 to be in saturation:

$V_{GS1} - V_{Th1} < V_{DS1}$

or

$V_{G1} - V_{Th1} < V_{D1}$

ie. $(V_{in} - V_{GS4}) - V_{Th1} < V_{DD} - |V_{ov}|_2 - |V_{ov}|_3$

This condition ~~may~~ violate for max^m $V_{in} = V_{DD}$

ie.

$(\cancel{V_{DD}} - V_{GS4}) - V_{Th1} < \cancel{V_{DD}} - |V_{ov}|_2 - |V_{ov}|_3$

or $V_{GS4} + V_{Th1} > |V_{ov}|_2 + |V_{ov}|_3$

$V_{GS4} > 0.2 + 0.2 - 0.3 = 0.1V$

(3M)

(ii) $V_{out(max)} = V_{DD} - |V_{ov}|_2 - |V_{ov}|_3$

$= 3.3 - 0.4 = 2.9V$

$V_{out(min)} = V_{OV(M1)}$

$= 0.2\ V$ (3M)

(iii) Max allowable dc voltage @ v (through M1)

$= V_{DD} - |V_{ov}|_2 - |V_{ov}|_3 - V_{Th1}$

$= 3.3 - 0.2 - 0.2 + 0.3$

$= 3.2\ V$

other way, $V_{in} - V_{GS} = 2.8$

$V_{DD} - 0.5 = 2.8$

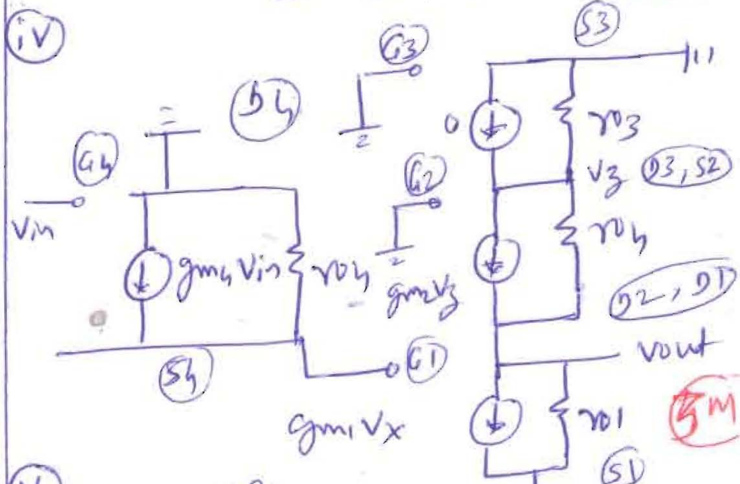
Max allowable dc voltage @ V_x

$= V_{DD} - |V_{ov}| = 3.3 - 0.2$

$= 3.1\ V$

∴ Max allowable dc voltage @ V_x = ~~3.1 V~~ 2.8V (3M)

(iv)



(v) $A_{v1} \cong \frac{r_{o4}}{r_{o4} + 1/g_{m4}} = \frac{25K}{25K + 500}$

$= 0.98\ V/V$

$A_{v2} \cong -g_{m1}[r_{o1} \| g_{m2} r_{o2} r_{o3}]$

$\cong -2 \times [25K \| 1250K]$

$\cong -2 \times 24.5 = -49\ V/V$

$A_v = A_{v1} \times A_{v2} = -48.03\ V/V$

(4M)

Q5 (b)

at Q_3.

$100\mu A = \frac{1}{2} \times 60 \frac{\mu A}{V^2} \times \left(\frac{40}{1}\right) \times (V_{GS} - 0.7)^2$

$V_{GS} = 0.99 \Rightarrow V_{ov} = 0.29$

$V_2 = 2.5 - 0.29 = \cancel{1.51}\ 2.21$

at o/p.

$1mA = \frac{1}{2} \times 120 \frac{\mu A}{V^2} \times \left(\frac{20}{1}\right) \times V_{ov}^2$

$V_{ov} = 0.91 \Rightarrow V_{GS} = V_{ov} + V_t = 1.61$ 1½

~~$V_{ov} = V_{GS} - V_{ov}$~~ $= 0.99 - 0.91 = 0.08$

(i.e. clos to 0)

$V_o = \underbrace{2.21}_{V_2} - \underbrace{1.61}_{V_{GS}} = 0.6.$ (i.e. close to 1) 1½

3

(a) Q_3 & Q_4 form Current Multiplier.

$\times \frac{120}{40} = \times 3$

$g_{m_2} = g_{m_1} = 2\sqrt{\frac{1}{2}(120) \times \left(\frac{20}{1}\right) \times 100} = 693 \mu A/V.$ ½+½

$g_{m_3} = 2\sqrt{\frac{1}{2} \times 60 \times \left(\frac{40}{1}\right) \times 100} = 693 \mu A/V$ ½

$g_{m_4} = 2\sqrt{\frac{1}{2} \times 60 \times \left(\frac{120}{1}\right) \times 300} = 2078 \mu A/V.$ ½

$g_{m_5} = 2\sqrt{\frac{1}{2} \times 60 \times \left(\frac{20}{1}\right) \times 1000} = 1550 \mu A/V$ ½

$r_{o_1} = r_{o_2} = r_{o_3} = \frac{24}{100} = 240 k\Omega$ ½

$r_{o4} = 24/300 = 80 k\Omega$ ½

$r_{o5} = 24/1000 = 24 k$ ½

(5)

3+3

(c) Open loop gain

$A\beta \simeq g_{m_1}(r_{o_1} \| r_{o}$

$\left[(3 \times g_{m_1})\left(\frac{r_{o4}}{3}\right) \equiv g_{m_1} r_{o}\right.$

$\boxed{\beta = 1}$

$\therefore A = g_{m_1}(r_{o_2} \| r_{o_3}$

$\boxed{A \simeq 693 \times 120 \times 10^{-3} =}$

(d) $A_f = \frac{A}{1 + A\beta} = \frac{84}{1 + 84 \times 1} = 0.9$

$R_o = r_{o5} \| r_{o5} = 12 k.$

$\boxed{R_{of} = \frac{R_o}{1 + A\beta} = \frac{12K}{1 + 84 \times 1} = 1}$

(e) To obtain $\frac{V_o}{V_S} = 5$

We could change direct connection from Q_{55} to Q_{26} by

Voltage divider $\frac{R_1}{R_1 + R_2}$ to

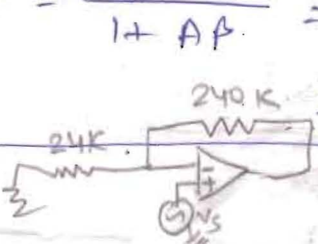
Change β from 1 to $\frac{1}{5.3}$

then.

$A\beta = \frac{84}{1 + 84 \times \frac{1}{5.3}} = \frac{84}{16.8}.$

New $1 + A\beta = 16.8.$

$\boxed{R_{of}'' = \frac{R_o}{1 + A\beta} = \frac{12}{16.8} = .714\ \Omega}$



5) (b)

Q3.

$100\mu A = \frac{1}{2} \times 60 \frac{\mu A}{V^2} \times \left(\frac{40}{1}\right) \times (V_{GS} - 0.7)^2$

$V_{GS} = 0.99 \Rightarrow V_{ov} = 0.29$

$V_2 = 2.5 - 0.29 = \cancel{1.51}\ 2.21$

Q1:

$1mA = \frac{1}{2} \times 120 \frac{\mu A}{V^2} \times \left(\frac{20}{1}\right) \times V_{ov}^2$

$V_{ov} = 0.91 \rightarrow V_{GS} = V_{ov} + V_t = 1.61$ ($1\frac{1}{2}$)

~~$V_o = V_{GS} - V_{ov} = 0.99 - 0.91 = 0.08$~~ (i.e. close to 0)

$V_o = \underbrace{2.21}_{V_2} - \underbrace{1.61}_{V_{GS}} = 0.6$. (i.e. close to 1) ($1\frac{1}{2}$)

Q3 & Q4 form Current Multiplier.

$\times \frac{120}{40} = \times 3$

$g_{m1} = 2\sqrt{\frac{1}{2}(120)\times\left(\frac{20}{1}\right)\times 100} = 693 \mu A/V$ ½

$= 2\sqrt{\frac{1}{2}\times 60\times\left(\frac{40}{1}\right)\times 100} = 693 \mu A/V$ ½

$= 2\sqrt{\frac{1}{2}\times 60\times\left(\frac{120}{1}\right)\times 300} = 2078 \mu A/V$ ½

$= 2\sqrt{\frac{1}{2}\times 60\times\left(\frac{20}{1}\right)\times 1000} = 1550 \mu A/V$ ½

$r_{o2} = r_{o3} = \frac{24}{100} = 240k\Omega$ ½

$24/300 = 80 k\Omega$ ½

$24/1000 = 24 k$ ½

3+3 (5)

(c) Open loop gain

$A\beta \cong g_{m1}(r_{o1} \| r_{o3}) \times 1$

$\left[(3\times g_{m1})\left(\frac{r_{o4}}{3}\right) \equiv g_{m1} \cdot r_{o1}\right]$.

$\boxed{\beta = 1}$

$\therefore A = g_{m1}(r_{o2} \| r_{o3})$

$\boxed{A \cong 693 \times 120 \times 10^{-3} = 84}$

(2)

(d) $A_f = \frac{A}{1+A\beta} = \frac{84}{1+84\times 1} = 0.988 \frac{V}{V}$ (1)

$R_o = r_{o5} \| r_{o5} = 12 k$.

$\boxed{R_{of} = \frac{R_o}{1+A\beta} = \frac{12K}{1+84\times 1} = 140\Omega}$ (1)

(e) To obtain $\frac{v_o}{v_s} = 5$

We could change direct connection from Q_{5S} to Q_{2G} by Voltage Divider $\frac{R_1}{R_1+R_2}$ to

Change β from 1 to $\frac{1}{5.3}$ then.

$A_f = \frac{84}{1+84\times\frac{1}{5.3}} = \frac{84}{16.8} = 5$

New $1 + A\beta = 16.8$.

(5)

$\boxed{R_{of}'' = \frac{R_o}{1+A\beta} = \frac{12}{16.8} = .714\ \Omega}$.